

ERIK BROCKBANK

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Last Updated: Spring, 2026

APPOINTMENTS

Stanford University Postdoctoral Fellow <i>Department of Psychology</i>	<i>Sep. 2023 -</i>
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EDUCATION

University of California, San Diego Ph.D. <i>Experimental Psychology</i>	<i>2023</i>
Stanford University B.S., M.S. <i>Symbolic Systems</i>	<i>2014</i>

ACADEMIC HONORS & AWARDS

PsychTerms Graduate Research Award	<i>2022</i>
Psychonomic Society Graduate Conference Award	<i>2020</i>
Cognitive Science Society Computational Prize	<i>2020</i>

FELLOWSHIPS & GRANTS

NSF SBE Postdoctoral Research Fellowship	<i>2024-2026</i>
UCSD Academic Senate General Research Grant	<i>2022</i>
UCSD Anderson Graduate Research Fellowship	<i>2019</i>
UCSD Katzin Graduate Fellowship	<i>2018-2023</i>

MANUSCRIPTS IN PREPARATION

Brockbank, E., Yang, J., Wang, H., Mirchandani, S., Bıyık, E., Sadigh, D., Gerstenberg, T., & Fan, J. E. (*in prep*). Impact of bias and variance on advice taking in intuitive physics domains.

JOURNAL ARTICLES

Brockbank, E., Verma, A., Lloyd, H., Huey, H., Padilla, L., & Fan, J. E. (2025). Evaluating convergence between two data visualization literacy assessments. *Cognitive Research: Principles and Implications*.

Brockbank, E., & Vul, E. (2024). Repeated rock, paper, scissors play reveals limits in adaptive sequential behavior. *Cognitive Psychology*.

Brockbank, E., Lombrozo, T., Gopnik, A., & Walker, C. (2022). Ask me why, don't tell me why: Asking children for explanations facilitates relational thinking. *Developmental Science*.

Brockbank, E., & Walker, C. (2022). Explanation Impacts Hypothesis Generation, but not Evaluation, During Learning. *Cognition*.

Brockbank, E., Vul, E. & Barner, D. (2022). Ongoing dynamic calibration produces unstable number estimates. *Journal of Experimental Psychology: General*.

Schneider, R., **Brockbank, E.,** Feiman, R., & Barner, D. (2021). Counting and the ontogenetic origins of exact equality. *Cognition*.

Brockbank, E., & Vul, E. (2021). Formalizing Opponent Modeling with the Rock, Paper, Scissors Game. *Games*.

Destefano, I., Oey, L.A., **Brockbank, E.**, & Vul, E. (2021). Integration by Parts: Collaboration and Topic Structure in the CogSci Community. *Topics in Cognitive Science*.

CONFERENCE PROCEEDINGS

Brockbank, E., Dee, N., O’Keeffe, M., Mahaphanit, W., Gerstenberg, T., Fan, J. E., & Hawkins, R. (2026). Asking the right questions? What people learn about strangers in conversation. In *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*.

Kuang, J., **Brockbank, E.**, Bicchieri, C., & Hawkins, R. (2026). Inferring Joint Commitment from Observed Coordination. In *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*.

*Cross, L., ***Brockbank, E.**, Gerstenberg, T., Fan, J. E., Yamins, D. L. K., & Haber, N. (2025). Understanding Human Limits in Pattern Recognition: A Computational Model of Sequential Reasoning in Rock, Paper, Scissors. In *Proceedings of the 8th Annual Conference on Cognitive Computational Neuroscience*.

Brockbank, E., Gerstenberg, T., Fan, J. E., & Hawkins, R. D. (2025). How do we get to know someone? Diagnostic questions for inferring personal traits. In *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

Teo, V., Wu, S., **Brockbank, E.**, & Gerstenberg, T. (2025). Leave a trace: Recursive reasoning about deceptive behavior. In *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

Zheng, K., **Brockbank, E.**, Schwartz, S., Yeager, D., Bryan, C., Dweck, C., & Fan, J. E. (2025). Linking student psychological orientation, engagement, and learning in college-level introductory data science. In *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

Jin, E., Huang, Z., Fränken, J. P., Liu, W., Cha, H., **Brockbank, E.**, Wu, S., Zhang, R., Wu, J., & Gerstenberg, T. (2024). MARPLE: A Benchmark for Long-Horizon Inference. In *NeurIPS Datasets & Benchmarks Track*.

***Brockbank, E.**, *Yang, J., Govil, M., Fan, J. E., & Gerstenberg, T. (2024). Without his cookies, he’s just a monster: A counterfactual simulation model of social explanation. In *Proceedings of the 46th Annual Conference of the Cognitive Science Society*.

*Wu, S., ***Brockbank, E.**, Cha, H., Fränken, J. P., Jin, E., Huang, Z., Liu, W., Zhang, R., Wu, J., & Gerstenberg, T. (2024). Whodunnit? Inferring what happened from multimodal evidence. In *Proceedings of the 46th Annual Conference of the Cognitive Science Society*.

Brockbank, E., Wang, H., Yang, J., Mirchandani, S., Bıyık, E., Sadigh, D., & Fan, J. (2022). How do people incorporate advice from artificial agents when making physical judgments? In *Proceedings of the 44th Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Vul, E. (2021). Humans fail to outwit adaptive rock, paper, scissors opponents. In *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Vul, E. (2020). Recursive Adversarial Reasoning in the Rock, Paper, Scissors Game. In *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Walker, C. (2020). Explanation Supports Hypothesis Generation in Learning. In *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*.

Oey, L.A., Destefano, I., **Brockbank, E.**, & Vul, E. (2020). Formalizing interdisciplinary collaboration in the CogSci community. In *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Vul, E. (2019). Mapping visual features onto numbers. In *Proceedings of the 41st Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Smith, K. & Vul, E. (2019). When do people use containment heuristics for physical predictions? In *Proceedings of the 41st Annual Conference of the Cognitive Science Society*. (Retraction November, 2019).

BOOK CHAPTERS

Brockbank, E., Holdaway, C., Acosta-Kane, D. & Vul, E. (2023). Sampling Data, Beliefs, and Actions. In Fiedler, K., Juslin, P., & Denrell, J. (Eds.), *Sampling in Judgment and Decision Making*. Cambridge University Press.

CONFERENCE PRESENTATIONS

Brockbank, E., Dee, N., O’Keeffe, M., Mahaphanit, W., Gerstenberg, T., Fan, J. E., & Hawkins, R. (2026). Asking the right questions? What people learn about strangers in conversation. Poster presented at the *48th Annual Conference of the Cognitive Science Society*.

Kuang, J., **Brockbank, E.**, Bicchieri, C., & Hawkins, R. (2026). Inferring Joint Commitment from Observed Coordination. Poster presented at the *48th Annual Conference of the Cognitive Science Society*.

*Cross, L., ***Brockbank, E.**, Gerstenberg, T., Fan, J. E., Yamins, D. L. K., & Haber, N. (2025). Understanding Human Limits in Pattern Recognition: A Computational Model of Sequential Reasoning in Rock, Paper, Scissors. Poster presented at the *8th Annual Conference on Cognitive Computational Neuroscience*.

Brockbank, E., Gerstenberg, T., Fan, J. E., & Hawkins, R. D. (2025). How do we get to know someone? Diagnostic questions for inferring personal traits. Poster presented at the *47th Annual Conference of the Cognitive Science Society*.

Teo, V., Wu, S., **Brockbank, E.**, & Gerstenberg, T. (2025). Leave a trace: Recursive reasoning about deceptive behavior. Poster presented at the *47th Annual Conference of the Cognitive Science Society*.

Zheng, K., **Brockbank, E.**, Schwartz, S., Yeager, D., Bryan, C., Dweck, C., & Fan, J. E. (2025). Linking student psychological orientation, engagement, and learning in college-level introductory data science. Poster presented at the *47th Annual Conference of the Cognitive Science Society*.

Hurwitz, E., **Brockbank, E.**, & Walker, C. M. (2025). Contexto: A New Approach to Measure Flexible Reasoning Under Dynamic Constraints. Poster presented at the *47th Annual Conference of the Cognitive Science Society*.

Brockbank, E., Gerstenberg, T., Fan, J. E., & Hawkins, R. D. (2025). How do we get to know someone? Diagnostic questions for inferring personal traits. Poster presented at the *51st Annual Conference of the Society for Philosophy and Psychology*.

Jin, E., Huang, Z., Fränken, J. P., Liu, W., Cha, H., **Brockbank, E.**, Wu, S., Zhang, R., Wu, J., & Gerstenberg, T. (2024). MARPLE: A Benchmark for Long-Horizon Inference. Poster presented at *NeurIPS Datasets & Benchmarks Track*.

***Brockbank, E.**, *Yang, J., Govil, M., Fan, J. E., & Gerstenberg, T. (2024). Without his cookies, he’s just a monster: A counterfactual simulation model of social explanation. Poster presented at the *46th Annual Conference of the Cognitive Science Society*.

*Wu, S., ***Brockbank, E.**, Cha, H., Fränken, J. P., Jin, E., Huang, Z., Liu, W., Zhang, R., Wu, J., & Gerstenberg, T. (2024). Whodunnit? Inferring what happened from multimodal evidence. Poster presented at the *46th Annual Conference of the Cognitive Science Society*.

***Brockbank, E.**, *Yang, J., Govil, M., Fan, J. E., & Gerstenberg, T. (2024). Without his cookies, he’s just a monster: A counterfactual simulation model of social explanation. Talk presented at the *50th Annual Conference of the Society for Philosophy and Psychology*.

Hurwitz, E., Dean, J., Farrell, B., Mckinty, A., **Brockbank, E.**, Gopnik, A., Walker, C. (2024). A Cognitive Development Model of the Effects of Psychedelics. Poster presented at the *50th Annual Conference of the Society for Philosophy and Psychology*.

Lloyd, H., Tait, Z., **Brockbank, E.**, Liao, S., & Fan, J. (2022). How well do student attitudes and engagement predict learning outcomes in introductory data science courses? Talk at the *PsychTerms Annual Meeting*.

Brockbank, E., Wang, H., Yang, J., Mirchandani, S., Bıyık, E., Sadigh, D., & Fan, J. (2022). How do people

incorporate advice from artificial agents when making physical judgments? Talk at the *44th Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Vul, E. (2021). Rock, paper, scissors agents exploit patterns in human sequential behavior. Talk at the *62nd Annual Meeting of the Psychonomic Society*.

Brockbank, E., & Vul, E. (2021). Humans fail to outwit adaptive rock, paper, scissors opponents. Poster presented at the *43rd Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Walker, C. (2021). Explanation Impacts Hypothesis Generation, but not Evaluation, during Learning. Poster presented at the *47th Annual Conference of the Society for Philosophy and Psychology*.

Brockbank, E., & Vul, E. (2020). Modeling Adaptive Reasoning in Rock, Paper, Scissors. Poster presented at the *61st Annual Meeting of the Psychonomic Society*.

Brockbank, E., & Vul, E. (2020). Recursive Adversarial Reasoning in the Rock, Paper, Scissors Game. Talk at the *42nd Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Walker, C. (2020). Explanation Supports Hypothesis Generation in Learning. Talk at the *42nd Annual Conference of the Cognitive Science Society*.

Oey, L.A., Destefano, I., **Brockbank, E., & Vul, E. (2020).** Formalizing interdisciplinary collaboration in the CogSci community. Talk at the *42nd Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Vul, E. (2020). Adaptive Reasoning in Rock-Paper-Scissors. Talk at the *53rd Annual Conference of the Society for Mathematical Psychology*.

Brockbank, E., & Vul, E. (2019). Mapping visual features onto numbers. Poster presented at the *41st Annual Conference of the Cognitive Science Society*.

Brockbank, E., Smith, K., & Vul, E. (2019). When do people use containment heuristics for physical predictions? Poster presented at the *41st Annual Conference of the Cognitive Science Society*.

Brockbank, E., & Vul, E. (2019). Mapping visual features onto numbers. Poster presented at the *45th Annual Conference of the Society for Philosophy and Psychology*.

INVITED TALKS

2026, March. Stanford University, Symbolic Systems Colloquium

2025, November. University of Tübingen, Germany, Linguistics Colloquium

2025, October. Berkeley Cognition Colloquium

2023, December. Stanford Social and Language Technology Reading Group

2022, October. Naval Research Laboratory “Reasoning Lunch Series”

TEACHING

Instructor of Record Data & Model Programming for Comp. Social Science	<i>Mar.-Jun. 2022</i>
Teaching Assistant Research Methods in Psychology	<i>Jan.-Mar. 2023</i>
Teaching Assistant Research Methods in Psychology	<i>Sep.-Dec. 2022</i>
Teaching Assistant Research Methods in Psychology	<i>Sep.-Dec. 2021</i>
Teaching Assistant Computational Social Science Foundations (python)	<i>Aug.-Sep. 2021</i>
Teaching Assistant Psychology: Cognitive Foundations	<i>Mar.-Jun. 2021</i>
Teaching Assistant Introduction to Psychology	<i>Sep.-Dec. 2020</i>
Teaching Assistant Psychology of Parenting	<i>Mar.-Jun. 2020</i>

Teaching Assistant Introduction to Psychology	<i>Sep.-Dec. 2019</i>
Teaching Assistant Industrial & Organizational Psychology	<i>Jul.-Aug. 2019</i>
Teaching Assistant Developmental Psychology	<i>Jan.-Mar. 2019</i>

MENTORSHIP

Nora Dee, Stanford Undergraduate Honors Thesis	<i>Jun. 2025-</i>
Misha O'Keefe, Stanford Research Staff	<i>Jun. 2025-</i>
Yiheng Yao, Stanford Master's Thesis	<i>Mar. 2025-</i>
Verona Teo, Stanford Research Staff	<i>Aug. 2024-</i>
Kristine Zheng, Stanford Research Staff	<i>Jul. 2024-Jul. 2025</i>
Mishika Govil, Stanford Undergraduate Researcher	<i>Nov. 2023-Jun. 2025</i>
Alison Yu, UCSD Undergraduate Researcher	<i>Apr. 2021-Mar. 2023</i>
MJ Mei, UCSD Undergraduate Researcher	<i>Apr. 2021-Mar. 2023</i>
Ziling Gong, UCSD Undergraduate Researcher	<i>Apr. 2021-Mar. 2023</i>
Annalea O'Halloran, UCSD Undergraduate Researcher	<i>Apr. 2021-Dec. 2021</i>

LEADERSHIP & OUTREACH

<i>UCSD Graduate Student Association</i> Psychology Dept. Representative	<i>2022-2023</i>
<i>UCSD Psychology Department</i> Climate Committee Member	<i>2020-2023</i>
<i>UCSD Psychology Department</i> Graduate Statistics Advisor	<i>2019-2023</i>
<i>UCSD Psychology Student Speaker Series</i> Graduate Student Officer	<i>2019-2023</i>
<i>UCSD Graduate Students</i> GradAMP Mentor	<i>2020-2021</i>
<i>UCSD Psychology Data Science Club</i> Club Leader	<i>2019-2020</i>
<i>UCSD Psychology Colloquium Series</i> Graduate Student Officer	<i>2019-2020</i>

PROFESSIONAL SERVICE

Ad hoc reviewer: *Child Development, Cognition, Cognitive Science, Journal of Experimental Psychology: General, Journal of Experimental Psychology: Learning, Memory, and Cognition, Thinking & Reasoning, Annual Meeting of the Cognitive Science Society, Annual Meeting of the Society for Philosophy and Psychology*